Week 9 Homework

Exercise

From the text, Exercise 3.2.

Exercise

From the text, Exercise 3.11.

Exercise

Find the definition for **isomorphic** graphs. What is the definition? Give an example of two graphs that are isomorphic, and give an example of two graphs that are not isomorphic.

Exercise

A simple graph G is **self-complimentary** if G and $\bar{G}$ are isomorphic. Find a self-complimentary graph having five vertices. Find another self complimentary graph with more than 1 vertex.

Exercise

A vertex v in a connected graph G is an **articulation point** if the removal of v and all edges incident to v disconnects G. Give an example of a graph with six vertices that has exactly two articulation points. Give an example of a graph on six vertices that has no articulation points.

Proof

Prove that if G is a simple graph with p vertices, where each vertex has degree $\geq \frac{1}{2}(p-1)$, then G must be connected. Hint: Try a contradiction proof, by assuming there are at least two components of G. How many vertices must each component have?

Proof

Prove the statement: In any tree there is exactly one path from any vertex to any other vertex. (Hint: Try contradiction. If there are two paths try to show there is a cycle.)

Proof

A **forest** is a simple graph where each component is a tree. Find a formula for the number of edges in a forest with n vertices and c components. Prove your formula.

Proof

Prove the average degree in a tree is always less than 2. More specifically express this average as a function of the number of vertices in tree.

Proof

Let v and w be distinct vertices in the complete graph K_n for $n \geq 2$. Prove the number of paths from v to w is

$$(n-2)!\sum_{k=0}^{n-2}\frac{1}{k!}$$

Proof

Prove that a vertex v in a connected graph G is an articulation point if and only if there are vertices w and x in G having the property that every path from w to x passes through v.

Proof

Prove that in any simple graph with more than one vertex, there must be two vertices with the same degree. (Hint: Try contradiction.)

Proof

Prove the number of vertices of degree 1 in an tree must be greater than or equal to the maximum degree in the tree.